

AC9M6N04 • Year 6 Mathematics

Adding and Subtracting Decimals

Use place value, efficient strategies, estimation and digital verification

We are learning to...

Students align place values, rename when regrouping, choose mental or written strategies and compare exact results with purposeful estimates.

Represent

Reason

Calculate

Interpret

Verify

Success criteria

- I can represent or identify the concept.
- I can explain the underlying relationship.
- I can select an appropriate strategy or feature.
- I can apply it in a new context.
- I can justify and verify the response.

Curriculum focus

apply knowledge of place value to add and subtract decimals, using digital tools where appropriate; use estimation and rounding to check the reasonableness of answers

Calculate $18.375 + 6.84$ by aligning place values

tens	ones	tenths	hundredths	thousandths
1	8	3	7	5
	6	8	4	0
2	5	2	1	5

Write 6.84 as 6.840 to align equal place values. Placeholder zeros clarify structure without changing value.

Choose an efficient subtraction strategy

$$32.00 - 9.875$$

written regrouping

$$19.95 + 7.60$$

compensation: $20 + 7.60 - 0.05$

$$48.6 - 19.9$$

subtract 20, then add 0.1

digital tool

verify after setup and estimation

Estimate before calculating: $18.375 + 6.84$ is about $18.4 + 6.8 = 25.2$, so 25.215 is reasonable.

Build and annotate the model

Represent adding and subtracting decimals and label the important parts, quantities or choices.

Compare and reason

Use the application model to compare two cases, explain the relationship and identify a likely error.

Transfer and verify

Apply the idea in an unfamiliar context and use a second method, evidence source or text feature to check it.

Common mix-ups

- **Decimal points not aligned:** Align place values, not final digits.
- **Zeros added to the left or inside incorrectly:** Only trailing placeholder zeros preserve a decimal's value.
- **Estimate omitted:** Use it to detect misplaced decimal points.
- **Calculator result accepted without context:** Check operation, input and magnitude.

Quick check

- Add 18.375 and 6.84.
- Subtract $32 - 9.875$.
- Use compensation for $48.6 - 19.9$.
- Estimate first.
- Explain placeholder zeros.

Teacher decision: continue when students explain the model and verify a new example; otherwise return to Slide 2.